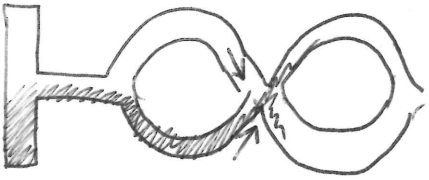
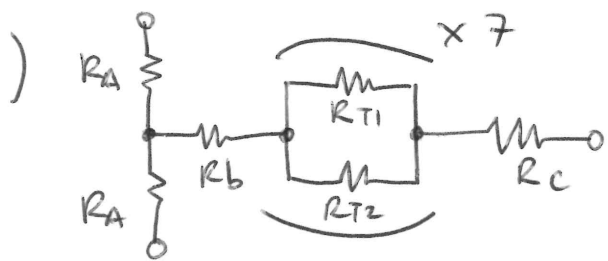
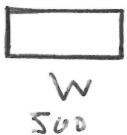


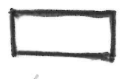
1)  Like in the Tesla mixer, laminar flow is broken by 

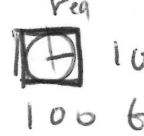
7 RINGS  $\rightarrow$  MIXING EFF.  $> 0.8$  OK

2) COMPACTNESS: GOOD  
POWER: MEDIUM (PUMP)  
EFFICIENCY: GOOD



$R_C = R_B \rightarrow$    $h=100$   $w=500$   $\alpha$

$R_{T1} \rightarrow$    $h=100$   $b_1=200$   $\beta$

$R_{T2} \rightarrow$    $h=100$   $b_2=100$   $\gamma$

$$\alpha = \frac{12\eta}{(1 - 0.63 \frac{1}{5}) (100 \cdot 10^{-6})^3 \cdot 500 \mu m} =$$

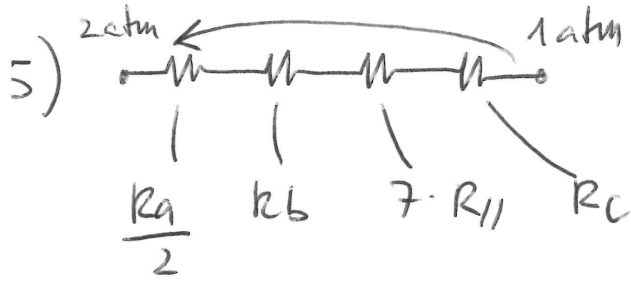
$$= \frac{12 \cdot 10^{-3}}{(100 \cdot 10^{-6})^3 \cdot 500 \cdot 10^{-6}} = 2.4 \cdot 10^{13}$$

$$\beta = \frac{12 \cdot \eta}{(1 - 0.63/2) \cdot (100 \cdot 10^{-6})^3 \cdot 200 \mu m} = 8.76 \cdot 10^{13}$$

$$\gamma = \frac{8 \cdot \eta}{\pi r_{eq}^4} = \frac{r_{eq} = \frac{2h^2}{4h} = h/2 = 50 \mu m}{\pi (50 \cdot 10^{-6})^4} = 4 \cdot 10^{14}$$

$$\left. \begin{aligned} Z_A &= \alpha \cdot 3 \text{ mm} = \frac{7.2 \cdot 10^{10}}{0.9} \\ Z_B &= \alpha \cdot 1 \text{ mm} = 2.4 \cdot 10^{10} = R_C \\ Z_{T1} &= \beta \cdot \frac{2\pi V}{2} = \frac{5.8 \cdot 10^{11}}{2} \cdot 2.75 \cdot 10^{11} \\ Z_{T2} &= \gamma \cdot \frac{2\pi V}{2} = \frac{2.5 \cdot 10^{12}}{2} \cdot 1.25 \cdot 10^{11} \end{aligned} \right\} \left[ \frac{\text{Pa} \cdot \text{s}}{\mu m^3} \right]$$

4)  $T \rightarrow q \rightarrow R$



$$Q = \frac{\Delta P}{R_{TOT}} = \frac{1 \text{ atm}}{R_{TOT}}$$

$$R_{||} = \frac{5 \cdot 5 \cdot 2.5}{(2.5 + 5.5)} = \frac{4.5 \cdot 10^{11}}{2} = 2.2 \cdot 10^{11}$$

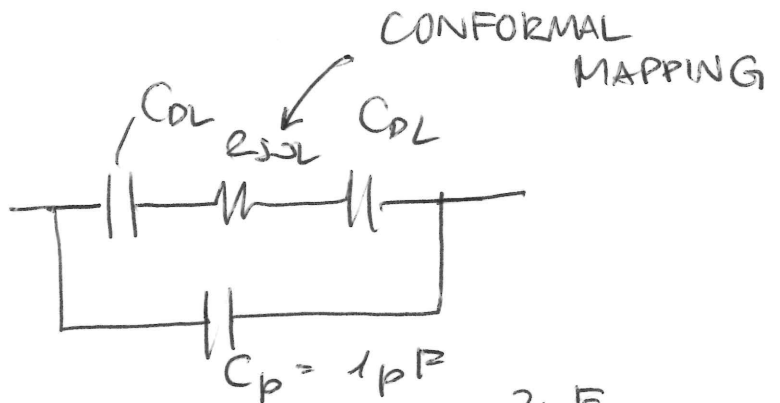
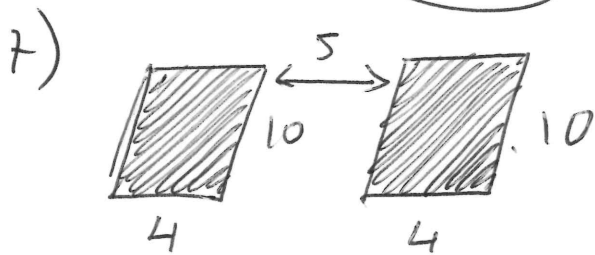
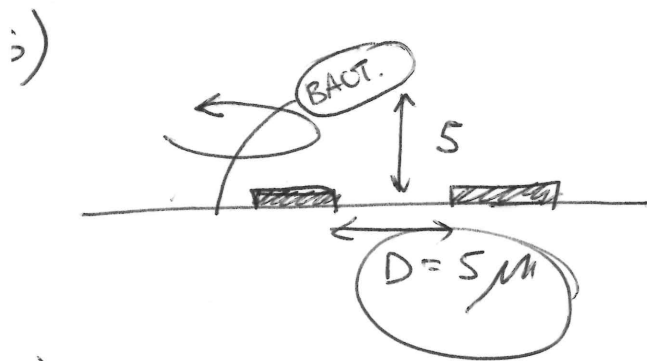
$$R_{TOT} = \frac{7.2}{2} \cdot 10^{10} + 2.4 \cdot 10^{10} + 7 \cdot \frac{4.5}{2} \cdot 10^{11} + 2.4 \cdot 10^{10} = 3.2 \cdot 10^{12}$$

6.2

$$Q = \frac{101 \text{ nPa}}{\frac{3.2 \cdot 10^{12} \text{ Pa} \cdot \text{s}}{1.65}} = 3.1 \cdot 10^{-8} \left[ \frac{\text{m}^3}{\text{s}} \right]$$

$$v = \frac{Q}{W \cdot h} = \frac{3.1 \cdot 10^{-8}}{100 \mu\text{m} \cdot 500 \mu\text{m}} = 0.62 \text{ m/s}$$

1.2 m/s



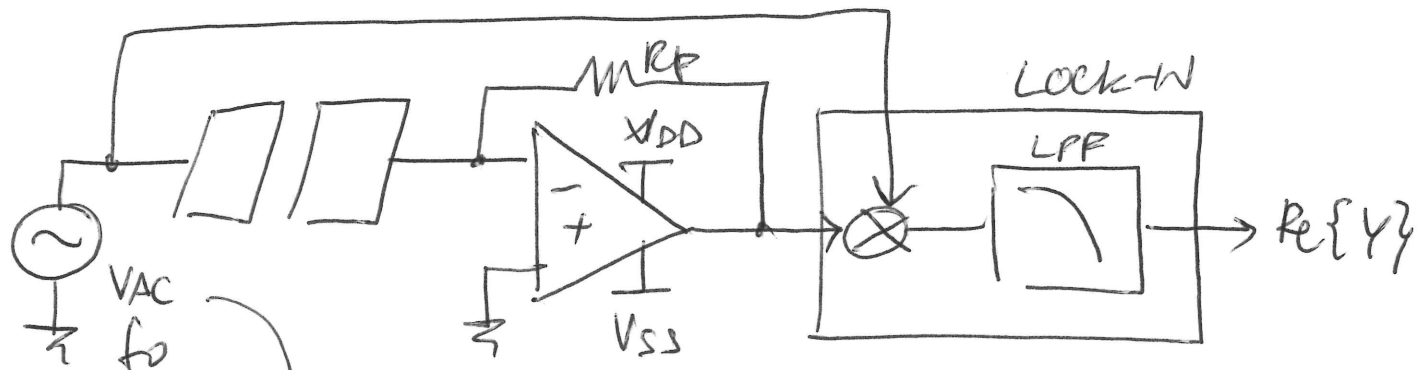
$$C_{DL} = 0.1 \frac{\text{pF}}{\mu\text{m}^2} \cdot 4 \cdot 10 \mu\text{m}^2 = 4 \text{ pF} \rightarrow$$

2 pF    $R_{sol}$    1 pF



NARROW  $R_{sol}$  PLATEAU

8)



$$V_{DD} = 5V$$

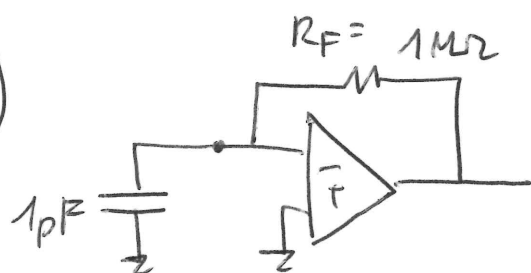
$$V_{SS} = -5V$$

$$I_{IN|_{max}} = 1\mu A \rightarrow$$

$$R_F = 1M\Omega$$

$$BW = \frac{1}{2\pi \cdot 1M\Omega \cdot 0.2pF} \approx 800kHz$$

9)



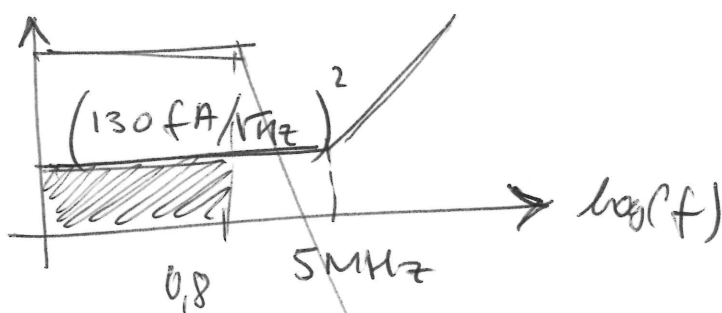
$$e_n = 4\mu V/\sqrt{Hz}$$

$$i_n = 2.5fA/\sqrt{Hz}$$

$$S_i|_{TOT} = i_n^2 + \frac{4kT}{R_F} + \frac{e_n^2}{(R_F)^2} + (2\pi f C_{TOT} \cdot e_n)^2 =$$

$$= (2.5 \cdot 10^{-15})^2 + \frac{4 \cdot 1.6 \cdot 10^{-19} \cdot 26mV}{10^6} + \left(\frac{4\mu V}{1M\Omega}\right)^2$$

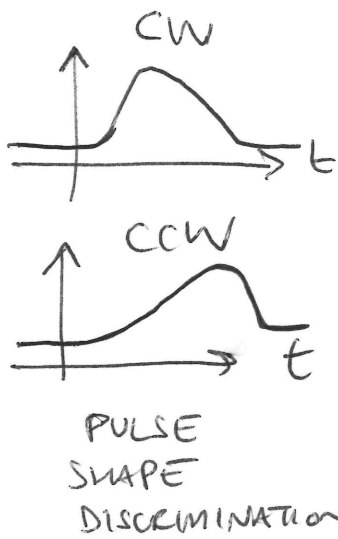
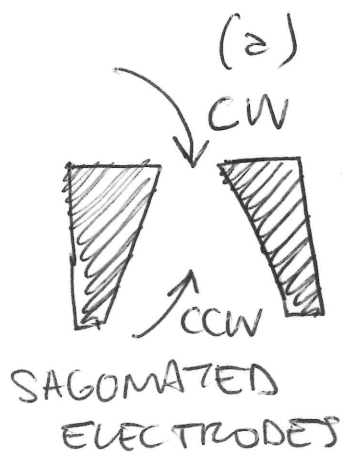
$$= (2.5 \cdot 10^{-15})^2 + (130 \cdot 10^{-15})^2 + (4 \cdot 10^{-15})^2 + \left(\frac{4\mu V}{1M\Omega} \cdot 2\pi f \cdot 1pF\right)^2$$



$$f_c = \frac{130 \cdot 10^{-15}}{4 \cdot 10^{-9} \cdot 2\pi \cdot 1pF} \approx 5MHz$$

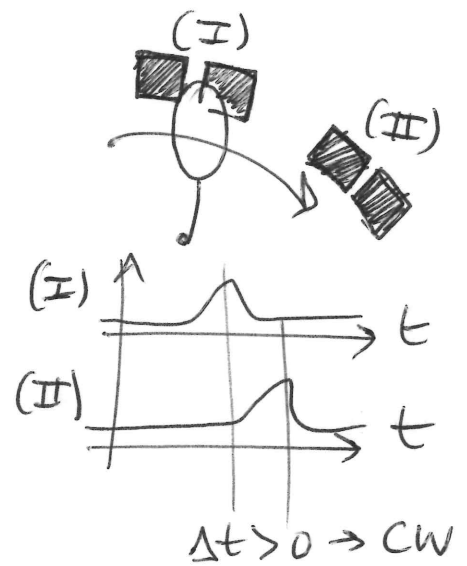
$$10) 10Hz \rightarrow LPF|_{BW} > 10Hz \sim \boxed{100Hz}$$

11)

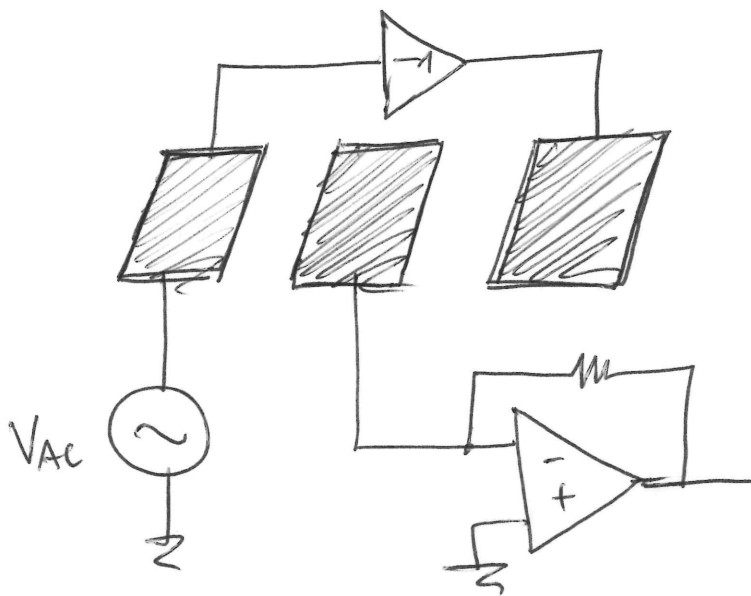


(b)

TWO DETECTOR



12)



OR

